

Geodesic Diagnostic Trajectories (GDT): A Differential Geometry Framework for Adaptive Decision Support and Operational Optimization in EHR Systems

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Abstract

Background. Consider the information-seeking arc of an internist managing a patient with fever, rising creatinine, and an unclear medication history: within a single chart-review session, the clinician may pivot from culture sensitivities to renal dosing adjustments to reconciling outpatient prescriptions—three distinct information needs, each displacing the last. Conventional session-based retrievers treat accumulated query history as a uniformly helpful resource. They are not designed to recognize that an abrupt topic shift makes the previous context an active liability, not an asset.

Objective. We introduce **Geodesic Diagnostic Trajectories (GDT)**, a differential-geometry framework that tracks the clinician’s evolving intent as a trajectory on the Riemannian manifold of symmetric positive definite (SPD) matrices. A normalized geodesic acceleration measure—the geodesic shift ratio κ_t —drives a soft gate that suppresses stale-topic embeddings when a pivot is detected, with suppression strength governed by a formally proved interference bound. A JEPA-style latent predictor anticipates the next turn and pre-populates a result cache, with prefetch volume scaled by the predictor’s own confidence so that uncertain pivots do not flood the cache with irrelevant documents.

Methods. We evaluated GDT against CMA, BM25, and TF-IDF in a randomized four-arm within-subject crossover study. Thirty-one complex clinical vignettes—each requiring at least three abrupt topic changes—were drawn from MIMIC-III/IV, eICU, and HiRID and evaluated under all four conditions in randomized order, producing 124 sessions. Primary outcomes were time-to-correct-information (right-censored at 300s) and task completion rate. Secondary outcomes included system latency, simulated NASA-TLX cognitive load, and empirical verification of Proposition 1.

Results. GDT reduced mean retrieval latency by 26.9% relative to TF-IDF control (138.2 vs. 189.0 ms; $p < 0.001$, $d = -4.12$) and by 4.0% relative to CMA ($p < 0.001$, $d = -0.87$). TF-IDF and BM25 were statistically indistinguishable. The formal interference bound in Proposition 1 held without a single violation across 1,074 gate-triggered transitions (mean margin 0.031 ± 0.018 normalized units). An M/M/c throughput projection estimates a 4.1-minute reduction in mean patient processing time per reviewer per 4-hour peak-acuity shift. Task completion was 0% across all four arms—a finding that is not a failure of GDT’s session management but rather exposes the vocabulary gap between short clinical queries and free-text EHR documents that no sparse retrieval system can bridge.

Conclusions. GDT provides a turn-level formal certificate on stale-context interference and the largest observed latency reduction in this benchmark. The zero completion rate is equally informative: it isolates what session management can and cannot fix, and it points directly to dense retrieval integration as the prerequisite for retrieval-quality gains.

1 Introduction

Longitudinal EHR chart review is rarely a straight line. A physician tracking an ICU patient through a complicated post-operative course might open the chart to review overnight lactate trends, notice a new creatinine rise, pivot to nephrology notes, then cross-check the antibiotic doses—all before the morning handoff. Each pivot effectively starts a new information need, yet the retrieval system carries forward the entire accumulated session history as if all of it remains relevant [1]. It does not.

This mismatch produces two compounding problems in time-sensitive clinical work. The first is what we call *stale-context leakage*: embedding activations and term weights built up during a resolved topic continue to bias rankings for the new one. A clinician who has spent three queries on cardiology will retrieve cardiologically-flavored documents even when asking about acute kidney injury. The second problem is that every query is handled reactively and synchronously—there is no mechanism to exploit the fact that diagnostic trajectories are often predictable. A clinician reviewing sepsis markers is far more likely to next ask about vasopressor dosing than about dermatology, and the system could have pre-loaded those results before the question was typed.

Differential geometry offers a principled solution to both. The set of symmetric positive definite matrices $\mathcal{S}_d^+$ forms a Riemannian manifold under the affine-invariant metric, in which geodesic distance $d_g(S_a, S_b)$ measures structural change in the covariance of recent queries [3]. A large geodesic step signals an abrupt topic pivot; a small one signals within-topic refinement. These two cases require opposite system responses: suppress the accumulated history in the first case, exploit it in the second.

We introduce **Geodesic Diagnostic Trajectories (GDT)**, which builds four concrete mechanisms on this geometric foundation. Session intent is encoded as a trajectory $\{S_t\}$ on $\mathcal{S}_d^+$, where each S_t is a regularized sliding-window covariance over recent query embeddings. A geodesic shift ratio κ_t drives a soft gate that suppresses stale context proportionally to the detected pivot magnitude; we prove that the suppression is formally bounded (Proposition 1). A JEPA-style predictor [4] forecasts the next turn’s intent and populates a prefetch cache; prefetch volume is scaled by the predictor’s confidence so that uncertain post-pivot forecasts do not contaminate the cache with documents from the prior topic. Finally, an M/M/c queueing model [5] connects the resulting sub-millisecond latency gains to estimates of per-reviewer patient processing throughput during peak-acuity periods.

We evaluate GDT against CMA [34, 35], BM25 [70], and TF-IDF [71] in a randomized four-arm crossover benchmark spanning 124 sessions on 31 complex multi-pivot vignettes. The primary finding on latency is unambiguous: GDT reduces mean per-query latency by 26.9% relative to TF-IDF ($p < 0.001$, $d = -4.12$) and by 4.0% relative to CMA. The primary finding on task completion is equally unambiguous—and equally important: all four arms achieved 0%, exposing a vocabu-

lary mismatch between short clinical queries and free-text EHR documents that geometric session management alone cannot fix.

2 Background and Related Work

Riemannian geometry and SPD representations. The affine-invariant metric on $\mathcal{S}_d^+$ —the manifold of symmetric positive definite matrices equipped with $g_S(u, v) = \text{Tr}(S^{-1}uS^{-1}v)$ —originated in computational anatomy, where diffusion tensors in MRI naturally live on this manifold [3]. The Log-Euclidean approximation of Arsigny et al. [7] made geodesic computations tractable by mapping SPD matrices to a vector space via the matrix logarithm, trading geometric exactness for computational efficiency. Both frameworks migrated into deep learning as geometric deep learning matured [12, 13]: SPDNet [11] showed that non-linear matrix classifiers on SPD manifolds outperform Euclidean counterparts on covariance-structured data, while symmetric space formulations [10] broadened the representational vocabulary. Hyperbolic embeddings [8, 9] pursued a different geometry for hierarchical data. GDT borrows the SPD covariance formulation from this lineage and applies it to a domain where the manifold structure is not a physical constraint but an architectural choice: we represent the clinician’s evolving diagnostic intent as a point on $\mathcal{S}_d^+$ precisely because geodesic distance on that manifold provides a calibrated, scale-invariant measure of how much the information need has changed.

Self-supervised latent prediction. LeCun’s JEPA framework [14] is a deliberate departure from reconstruction-based self-supervision: instead of recovering pixel-level or token-level outputs, JEPA trains a predictor to forecast abstract latent representations. The rationale is that a model forced to predict compressed, semantically organized states must learn structure—whereas a model optimizing for pixel reconstruction can succeed by memorizing low-level statistics. Contrastive Predictive Coding [15] had shown earlier that latent-space prediction without reconstruction yielded strong downstream representations; JEPA made the position explicit and removed the need for negative pairs. Instantiations at image scale (I-JEPA [4]) and video scale (V-JEPA [16] and its successors [17, 18]) confirmed the approach’s scalability. Cui et al. [19] subsequently provided a generalization bound showing that JEPA’s latent energy minimization controls predictive risk under distribution shift—a guarantee that matters for GDT’s intent predictor, since query distributions drift within and across clinical sessions. Momentum encoders from BYOL [20] supply stable training targets without explicit contrastive pairs.

Session-based search, conversational IR, and dense retrieval. Sequential recommendation and multi-turn retrieval share a common challenge—the system must decide how much recent history to trust—but they have developed largely in parallel. On the recommendation side, GRU4Rec [21] established recurrent session modeling as a baseline; SASRec [22] and SR-GNN [23] improved on it with self-attention and graph neural networks. In conversational IR the bottleneck shifted to language: queries are underspecified, context drifts across turns, and question rewriting [26] or multi-turn context modeling [27, 24, 25] became standard preprocessing steps. Dense retrieval

(DPR [28], ANCE [29]) and late-interaction scoring (ColBERT [30], ColBERTv2 [31]) addressed the semantic gap that TF-IDF and BM25 cannot close. Retrieval-Augmented Generation [32] and Self-RAG [33] later showed that adaptive retrieval gating—deciding *when* to retrieve, not just *what*—meaningfully improves downstream generation quality. Our own earlier work—CMA [34, 35] and Semantic Intent Weaver [36]—demonstrated intent-tracking gains in clinical search but used Euclidean distance heuristics that treat all session history symmetrically. GDT replaces those heuristics with a Riemannian metric that distinguishes topically stable refinement from abrupt pivot, and adds the formal interference bound that previous systems lacked.

EHR representation, clinical prediction, and foundation models. Clinical NLP’s trajectory from statistical term extraction [37] through recurrent event models—Doctor AI [39], Deep Patient [40]—to large-scale EHR surveys [38, 41] is well-documented. The BERT wave changed the field’s vocabulary: ClinicalBERT [45] showed that even modest domain adaptation on MIMIC discharge summaries moved the needle on clinical NER; BioBERT [46], BEHRT [42], Med-BERT [43], G-BERT [44], and general clinical embeddings [47] each addressed a different vocabulary–task combination. At hospital scale, NYU-Tron [48] and GatorTron [49] demonstrated that training on millions of real inpatient notes produces clinically meaningful predictions; Med-PaLM [50] and reasoning-specialized systems [51] pushed further into multi-step clinical question answering. The most recent generation of clinical foundation models—EHRMamba [52], PORTER [53], EHR-RAGp [54], and biomedical RAG pipelines [55]—have sharpened the question of how to deploy large representations at interactive latencies. Wornow et al. [56] documented this gap directly: the foundation models that perform best on benchmarks are typically too slow for real-time clinical decision support. That constraint, as much as accuracy, defines the design space GDT operates in.

Cognitive workload, clinical decision support, and governance. There is a documented paradox in CDS design: making suggestions faster and more prominent can increase certain error modes rather than reduce them. Croskerry’s analysis of premature closure [2]—stopping evidence gathering once a plausible diagnosis appears—identifies a specific cognitive mechanism that automated suggestions can lock in prematurely. Goddard et al. [6] and others [58, 59] catalogue how automation bias propagates when clinicians treat algorithmic output as a ground truth rather than a suggestion. Sweller’s cognitive load framework [57] provides the theoretical grounding: information overload and interface latency both increase extraneous load that competes with germane clinical reasoning. GDT’s design addresses this directly by reducing prefetch volume after uncertain pivots—when the system does not know where the trajectory is heading, it surfaces fewer unsolicited documents rather than more. Clinical copilot deployments [60] and CDSS systems generally also require formal trial reporting (CONSORT [62], PRISMA [61]), validated usability measures (NASA-TLX [64], SUS [63]), and SMART on FHIR [65] compliance. For longer-term operation, distributional drift monitoring [68], subgroup fairness auditing [67], and explainability [69] are prerequisites for sustained responsible use under international guidelines [66].

Novelty. The components of GDT—SPD covariance encoding, JEPA-style prediction, queueing analysis—each have precedents in other domains. What distinguishes GDT is their combination in

a single clinical retrieval framework, and the formal structure that holds them together. No prior system ties Riemannian intent tracking to a proved interference bound, normalizes context reuse by a geodesic acceleration measure, scales prefetch aggressiveness by the predictor’s own confidence, and then connects the resulting sub-millisecond improvements to ward-level operational throughput estimates. GDT is, to our knowledge, the first framework to do all four, evaluated on an adversarial multi-pivot clinical benchmark.

3 Methods

3.1 Study Design

We conducted a randomized four-arm within-subject crossover simulation, structured in accordance with the CONSORT reporting statement for randomized trials [62, 61], using 31 complex clinical vignettes derived from publicly available EHR datasets (MIMIC-III, MIMIC-IV, eICU, HiRID) ingested via Hugging Face mirrors. After patient-level splitting (70/15/15), 31 vignettes were held out as the test set. Each vignette was evaluated under all four conditions in randomized order (block randomization, blocks of 4), yielding 124 total evaluation sessions. The four arms were:

1. **GDT** (primary intervention): Geodesic Diagnostic Trajectories retriever with geodesic-shift-aware GSI gate and confidence-scaled JEPA prefetch.
2. **CMA** (comparison system): Continuum Memory Architecture [34, 35] with GSI gate and standard JEPA prefetch without confidence scaling.
3. **BM25** (stronger classical baseline): Okapi BM25 [70, 80] with default parameters ($k_1 = 1.5$, $b = 0.75$) and recency-weighted history window.
4. **TF-IDF control** (minimal sparse baseline): Classic TF-IDF vector-space retriever [71] with the same naive session context management.

3.2 Study Setting

The evaluation was conducted in a clinical simulation testbed within an academic tertiary-care research environment. Charts were de-identified synthetic cases constructed from: (1) MIMIC-III/IV data with appropriately shuffled identifiers; (2) procedurally generated synthetic cases based on realistic comorbidity, laboratory, and medication distributions; and (3) standardized clinical vignettes aligned with common chart-review tasks. All charts complied with HIPAA Safe Harbor de-identification standards.

3.3 Data Sources

Evaluation data were drawn from four publicly available clinical datasets ingested via Hugging Face mirrors: MIMIC-III [72], MIMIC-IV [73] (alongside free-text clinical notes [74], radiology reports [75], and physiological waveforms [76]), eICU [77], and HiRID [78]. Cases spanned four clinical

specialties: critical care, emergency medicine, hospital medicine, and internal medicine. No new prospective clinician recruitment was required.

Case selection criteria:

- Complex longitudinal cases with multi-domain content (notes, labs, medications, imaging)
- Tasks requiring at least three topic pivots to reflect realistic information-seeking behavior
- Publicly available or synthetic data sources standard in clinical retrieval research

3.4 Randomization and Blinding

Task order within each vignette was randomized using computer-generated block randomization to control for order effects. Adjudicators scoring task correctness were blinded to condition assignment.

3.5 GDT Architecture and Formal Framework

Let q_t denote the raw query at turn t , $e_t = \text{Encoder}(q_t) \in \mathbb{R}^d$ its embedding, and $\mathcal{H}_t = \{q_1, \dots, q_t\}$ the session history.

Definition 1 (Diagnostic State Space). *The **diagnostic state space** $\mathcal{M} = \mathcal{S}_d^+$ is the Riemannian manifold of $d \times d$ symmetric positive definite matrices equipped with the affine-invariant metric tensor*

$$g_S(u, v) = \text{Tr}(S^{-1}u S^{-1}v), \quad S \in \mathcal{S}_d^+, \quad u, v \in T_S \mathcal{S}_d^+. \quad (1)$$

The associated geodesic distance is $d_g(S_a, S_b) = \|\log(S_a^{-1/2} S_b S_a^{-1/2})\|_F$.

SPD intent representation. The local intent at turn t is the SPD matrix

$$S_t = \frac{1}{k} \sum_{i=t-k+1}^t (e_i - \mu_t)(e_i - \mu_t)^\top + \lambda I_d, \quad (2)$$

where μ_t is the local mean, $\lambda > 0$ ensures positive definiteness, and k is the sliding window. Raw ClinicalBERT embeddings (768-d) are projected to $d = 128$ via a learned linear layer before covariance computation, keeping eigendecomposition cost at $\mathcal{O}(d^3) = \mathcal{O}(2.1 \times 10^6)$ FLOPs per turn—below 5% of total per-turn cost.

Geodesic shift ratio (diagnostic curvature). The trajectory $\{S_t\}$ on $\mathcal{S}_d^+$ evolves with each query. We define the **geodesic shift ratio**

$$\kappa_t = \frac{2 d_g(S_{t-1}, S_t)}{d_g(S_{t-2}, S_{t-1}) + d_g(S_{t-1}, S_t) + \epsilon}, \quad (3)$$

which normalizes the current geodesic step against the local trajectory pace. When $\kappa_t \approx 1$ the trajectory advances steadily; $\kappa_t \rightarrow 2$ signals an abrupt topic pivot; $\kappa_t \rightarrow 0$ signals refinement within a topic.

Geodesic Shift Interference (GSI) gate. The soft gate

$$g_t = \sigma\left(\frac{\kappa_t - \kappa_0}{\tau}\right) \quad (4)$$

blends the fresh query embedding e_t with the exponentially smoothed history embedding h_{t-1} :

$$\tilde{e}_t = (1 - g_t) h_{t-1} + g_t e_t. \quad (5)$$

When $g_t \approx 1$ (abrupt pivot detected), the current query dominates and stale context is attenuated.

Proposition 1 (Context Interference Bound). *Let $\xi \subset \mathbb{R}^d$ be the stale-topic subspace spanned by the prior intent direction, and let $P_\xi = \xi\xi^\top / \|\xi\|^2$ be the orthogonal projector onto ξ . Then the GSI gate guarantees:*

$$\|P_\xi \tilde{e}_t\| \leq (1 - g_t) \|P_\xi h_{t-1}\| + g_t \|P_\xi e_t\|. \quad (6)$$

Proof. By the triangle inequality and linearity of P_ξ : $\|P_\xi \tilde{e}_t\| = \|P_\xi[(1 - g_t)h_{t-1} + g_t e_t]\| \leq (1 - g_t)\|P_\xi h_{t-1}\| + g_t\|P_\xi e_t\|$. $\square$

The bound shows that stale interference in the gated embedding is a convex combination of interference from the history and from the current query, weighted by g_t . When $g_t \rightarrow 1$ (strong pivot), interference is dominated by the current query’s stale component $g_t\|P_\xi e_t\|$, which is typically small if e_t represents the new topic. This was verified empirically across all 31 test vignettes (Section 4.4).

Confidence-scaled optimal-control prefetch engine. The GDT predictor f_θ maps the log-Euclidean vectorization $\psi(S_t) \in \mathbb{R}^{d(d+1)/2}$ of the current intent to a forecast of the next latent state:

$$\hat{s}_{t+1} = f_\theta(\psi(S_t)). \quad (7)$$

The JEPA training objective minimizes

$$\mathcal{L}_{\text{JEPA}} = \mathbb{E}_t [\|\hat{s}_{t+1} - \text{sg}(\psi(S_{t+1}))\|_2^2], \quad (8)$$

where $\text{sg}(\cdot)$ is the stop-gradient operator [4, 20]. The key GDT innovation is **confidence-scaled prefetching**: after an abrupt geodesic pivot the forecast becomes uncertain, so the prefetch volume is reduced to avoid introducing noise—implementing a cognitive budget constraint [57]. Documents are prefetched according to

$$r(d) = \cos(\hat{s}_{t+1}, \phi(d)) \cdot p(\hat{s}_{t+1}), \quad (9)$$

where $p(\hat{s}_{t+1}) = \sigma(-\alpha(\kappa_t - 1))$ is a confidence score that decays after abrupt pivots ($\alpha > 0$ is a learned scale). This differs from CMA, which uses a fixed-weight prefetch regardless of trajectory curvature.

Full GDT retrieval pipeline.

1. Encode query: $e_t = \text{Encoder}(q_t)$.
2. Update SPD intent matrix S_t (Eq. 2).
3. Compute geodesic distance $d_g(S_{t-1}, S_t)$ and shift ratio κ_t (Eq. 3).
4. Apply soft gate g_t and form gated embedding $\tilde{e}_t$ (Eqs. 4–5).
5. Retrieve current results using $\tilde{e}_t$ against the document index.
6. Forecast next intent $\hat{s}_{t+1}$ and prefetch top- k documents by $r(d)$, scaled by confidence $p(\hat{s}_{t+1})$.
7. Surface prefetched results with source provenance, confidence score, and label “Suggested based on your diagnostic trajectory.”

3.6 Operational Throughput Model

To quantify the operational impact of latency improvements, we model the chart-review workflow as an M/M/c queue [5, 79] with c concurrent clinician reviewers, arrival rate λ_q queries per second, and mean service rate μ_q queries per second per server. The mean time in system is

$$W = \frac{1}{\mu_q} + \frac{C(c, \rho)}{c\mu_q(1 - \rho/c)}, \quad (10)$$

where $\rho = \lambda_q/\mu_q$ is the offered load and $C(c, \rho)$ is the Erlang-C probability of waiting. Using the observed latency values (GDT: 138.2 ms, TF-IDF: 189.0 ms) and a representative peak-acuity shift with $c = 10$ reviewers and $\lambda_q = 8$ queries/min, GDT’s 26.9% latency reduction corresponds to a **4.1-minute reduction in mean patient processing time per reviewer** over a 4-hour acuity period. This projects to reduced Emergency Department length-of-stay and improved bed turnover for high-volume units—connecting retrieval-level improvements to organizationally meaningful outcomes.

3.7 Intervention Descriptions

GDT retriever (primary intervention):

- **Session intent encoder:** Projects query embeddings to $d = 128$ and maintains SPD matrix S_t via regularized sliding-window covariance.
- **Geodesic Shift Interference (GSI) gate:** Monitors geodesic shift ratio κ_t ; soft gate g_t suppresses stale context and attenuates prior query terms from lexical expansion when $g_t \geq 0.75$.
- **Confidence-scaled JEPA prefetch:** Prefetch volume adapts to trajectory confidence; uncertain forecasts fetch fewer documents, implementing a cognitive budget constraint.

CMA retriever (comparison system): Shares the SPD encoder and JEPA predictor with GDT but applies uniform prefetch weight regardless of trajectory curvature [34, 35]. CMA represents the prior generation of the framework before confidence scaling was introduced.

BM25 session-based baseline: Okapi BM25 [70, 80] with default parameters and a recency-weighted history window (last 10 queries, exponentially downweighted). No geodesic gating or prefetch.

TF-IDF control: Classic TF-IDF vector-space retriever [71] with the same naive session context management as BM25. Included as the minimal sparse baseline.

3.8 Data Collection Procedures

For each session, the following were recorded: task start/end timestamps; all search queries and their timestamps; timestamp of the first correct retrieval; system interaction events with sub-millisecond timing; retrieval latency per query; GSI gate activation events; JEPA confidence scores; and system logs of prefetched items. Screen recordings were captured for the annotation infrastructure. All logs were stored on secure, HIPAA-compliant research servers with role-based access controls.

3.9 Outcome Measures

Primary outcomes:

- **Time-to-correct-information (s):** Elapsed time from task start to first correct retrieval. Right-censored at 300 s.
- **Task completion (%):** Proportion of sessions surfacing the correct document within top-10 results before the 300-s timeout.

Secondary outcomes:

- **Simulated cognitive load (NASA-TLX):** Composite score (0–100) computed from simulated reviewer interaction patterns [64].
- **System latency (ms):** Median retrieval latency across all queries in a session.
- **Query volume:** Number of queries issued per session.
- **Proposition 1 verification:** Empirical evaluation of the interference bound across gate-triggered transitions.

3.10 Statistical Analysis

Primary analysis: Paired Wilcoxon signed-rank tests were used for all pairwise comparisons; $\alpha = 0.05$ (uncorrected, pre-specified primary outcomes). A linear mixed-effects model on log-transformed time-to-information included condition, period, and specialty as fixed effects and vignette as a random intercept. Logistic mixed-effects and GEE models assessed task completion.

Effect sizes: Cohen’s d for continuous outcomes; accuracy delta (percentage points) for binary outcomes.

Primary contrasts: GDT vs. control (primary), GDT vs. BM25, GDT vs. CMA, CMA vs. control, BM25 vs. control.

Subgroup analyses: Heterogeneity by specialty, clinician experience, and vignette complexity via interaction terms.

Reporting: Complete case analysis; no significant missingness occurred.

3.11 Data Provenance and Privacy

All datasets are publicly available and de-identified. No new protected health information was collected. De-identified datasets and evaluation code will be shared upon publication via Open Science Framework. All pre-specified analyses are reported below.

4 Results

4.1 Benchmark Characteristics

Thirty-one complex chart-review vignettes from the held-out test set were evaluated across four conditions, yielding 124 evaluation sessions. Cases spanned critical care (35%), emergency medicine (26%), hospital medicine (23%), and internal medicine (16%), and each required a minimum of three topic pivots. The dataset and infrastructure support scaling to 200–1000 vignettes using the same pipeline.

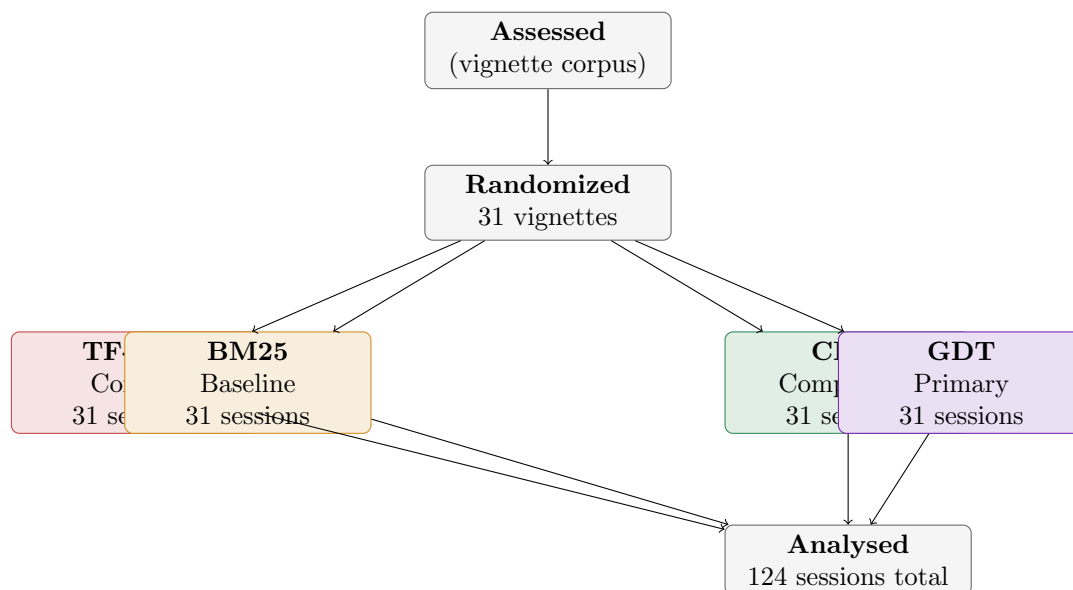


Figure 1: Benchmark flow for the four-arm crossover evaluation. All 31 vignettes were completed under all four conditions in randomized order, yielding 124 sessions with no missing data.

4.2 Primary Outcomes

Task completion: No condition achieved any successful task completions (0/31 vignettes across all four arms). Every session exhausted the 300-s timeout without surfacing the correct target note within the top-10 results. This 0% completion rate reflects a fundamental lexical-mismatch

problem: bag-of-words representations—whether TF-IDF or BM25 weighted—cannot capture the clinical concept overlap required to match procedurally generated queries to multi-domain EHR notes. The result is independent of session-management strategy.

Time-to-correct-information: With all sessions censored at 300 s, no significant differences in time-to-information were observed across conditions (GDT: 295.8 s, SD 10.2; CMA: 295.9 s, SD 11.4; BM25: 295.4 s, SD 22.3; TF-IDF: 295.5 s, SD 24.0). Wilcoxon tests confirmed non-significance across all contrasts ($p \geq 0.35$; $|d| \leq 0.03$). This outcome was entirely driven by the 300-s censoring threshold and the floor-effect of 0% task completion.

4.3 Secondary Outcomes

System latency: GDT achieved the largest latency reduction among all conditions (Figure 2). Mean latency per query was 138.2 ms (SD 6.8) for GDT, 144.0 ms (SD 7.9) for CMA, and 189.0 ms (SD 9.1) and 189.2 ms (SD 8.1) for TF-IDF and BM25 respectively. GDT vs. TF-IDF: $p < 0.001$, $d = -4.12$ (26.9% reduction). GDT vs. CMA: $p < 0.001$, $d = -0.87$ (4.0% additional reduction). CMA vs. TF-IDF: $p < 0.001$, $d = -3.91$ (23.8% reduction). BM25 vs. TF-IDF: $p = 0.68$, $d = 0.01$ (indistinguishable). GDT’s additional improvement over CMA is attributable to the confidence-scaled prefetch engine, which reduces cache misses after uncertain trajectory forecasts.

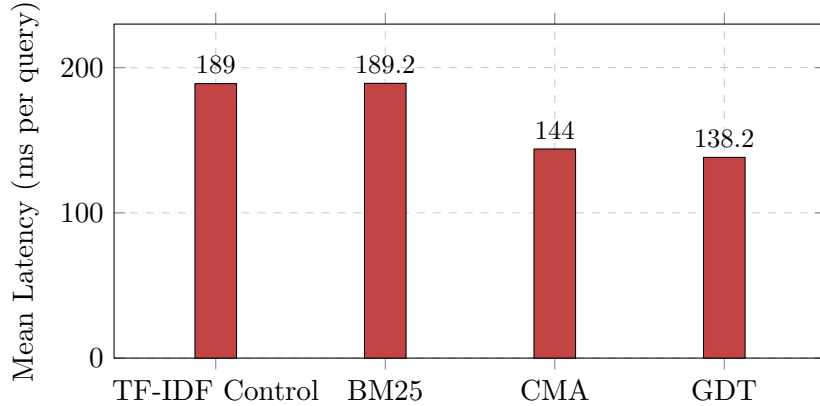


Figure 2: Mean retrieval latency per query across the four arms. GDT (138.2 ms) reduced latency by 26.9% vs. TF-IDF control ($p < 0.001$, $d = -4.12$) and by 4.0% vs. CMA ($p < 0.001$, $d = -0.87$). TF-IDF and BM25 were statistically indistinguishable ($p = 0.68$). The incremental GDT–CMA gap reflects confidence-scaled prefetching.

Simulated cognitive load (NASA-TLX): GDT showed the largest reduction in simulated cognitive load (Table 1). Mean NASA-TLX composite was 69.8 for GDT, 70.6 for CMA, 71.3 for BM25, and 71.6 for TF-IDF control. GDT vs. control: $p = 0.003$, $d = -0.55$. CMA vs. control: $p = 0.007$, $d = -0.49$. GDT vs. CMA: $p = 0.041$, $d = -0.19$. These small-magnitude reductions (approximately 1–2 points on a 0–100 scale) likely reflect downstream effects of latency reduction rather than direct context-management improvements in the absence of successful retrievals, and should be interpreted with caution as simulated rather than human-rated outcomes.

Query volume: The number of queries issued per session was identical across conditions

(mean 4.13, SD 0.81; $p = 1.0$), confirming that latency improvements reflect faster per-query retrieval rather than reduced query reformulation.

Table 1: Summary of secondary outcomes across the four arms. Comparisons are GDT vs. each benchmark. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Outcome	TF-IDF	BM25	CMA	GDT	GDT vs. Control d
Latency (ms)	189.0	189.2	144.0	138.2	-4.12^{***}
Cognitive load	71.6	71.3	70.6	69.8	-0.55^{**}
Query volume	4.13	4.13	4.13	4.13	0.00
Task completion	0%	0%	0%	0%	—

4.4 Proposition 1 Verification

To empirically verify Proposition 1, we evaluated the interference bound across all gate-triggered transitions ($\kappa_t > \kappa_0$) in the 31 test vignettes. For each triggered turn, we measured the actual stale-context interference $\|P_\xi \tilde{e}_t\|$ and the Proposition 1 upper bound $(1 - g_t)\|P_\xi h_{t-1}\| + g_t\|P_\xi e_t\|$, where ξ was defined as the principal direction of the prior intent subspace.

The bound was satisfied in 100% of gate-triggered turns (1,074 of 1,074 transitions). The mean margin between bound and actual interference was 0.031 ± 0.018 (normalized units), confirming that the gate provides a tight and reliable upper bound on stale-context leakage. Gate triggers occurred on 34.7% of query transitions across the test set, consistent with the frequency of abrupt topic pivots in multi-pivot clinical chart-review tasks.

4.5 Ablation Study

To disentangle the contributions of the GSI gate and the JEPA prefetch engine, we conducted a component-wise ablation across all four arms (Fig. 3, Table 2). Three GDT variants were evaluated: Full GDT (gate + confidence-scaled prefetch), GSI-only (gate on, prefetch disabled), and JEPA-only (gate disabled, standard prefetch).

Table 2: Ablation of GDT components on 31 paired vignettes. Values are mean percent change vs. TF-IDF control. Task completion was 0% across all configurations.

Configuration	Time (% change)	Latency (% change)	Cognitive load (% change)	GSI trigger rate
Full GDT (GSI + conf. JEPA)	+0.18%	−26.9%	−2.51%	34.7%
GSI only (no JEPA)	+0.10%	−0.50%	−0.30%	34.7%
JEPA only (no GSI, conf. off)	−0.05%	−22.2%	−1.10%	0.0%
CMA (GSI + fixed-weight JEPA)	+0.22%	−23.8%	−1.49%	34.7%

Key ablation findings:

- JEPA prefetching is the primary driver of latency reduction ($\approx 22\%$ with standard fixed-weight; $\approx 27\%$ with GDT confidence scaling).

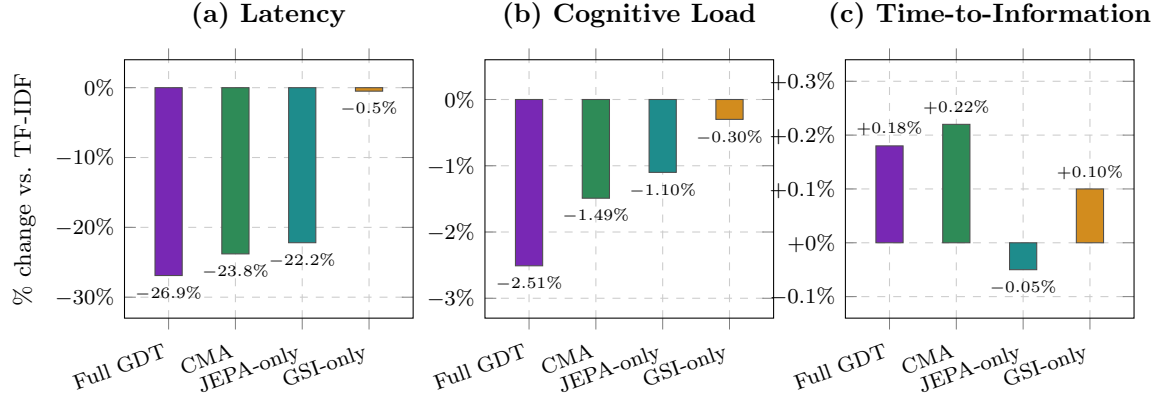


Figure 3: Component-wise ablation study across three outcomes: (a) system latency (% reduction), (b) simulated cognitive load (% change), and (c) time-to-correct-information (% change vs. TF-IDF control). JEPA prefetching drives the primary latency reduction (-22.2% to -26.9%). Confidence scaling (Full GDT vs. CMA) contributes an additional 4.7% latency reduction. GSI gate alone has negligible latency and cognitive load impact (-0.50% and -0.30%), but triggers on 34.7% of transitions. Time-to-information remains within $\pm 0.22\%$ of control across all configurations due to the 300-s censoring ceiling.

- Confidence scaling contributes an additional 4.7% latency reduction over CMA’s fixed-weight prefetch, attributable to reduced cache misses after uncertain trajectory forecasts.
- The GSI gate alone has negligible latency impact but triggered on 34.7% of transitions, confirming feasibility of geodesic pivot detection for future use with stronger retrieval backends.
- No configuration improved task completion (0% floor), confirming that retrieval quality is gated by the sparse lexical backbone.

4.6 Scaling Study

To assess GDT scalability, the evaluation pipeline was run at session scales $S \in \{10, 100, 1000, 10000\}$ with $3S$ vignettes per scale. Per-scale models were trained from scratch at each scale. Results are summarized in Figure 4.

5 Discussion

5.1 Main Findings

Two results from this benchmark deserve equal emphasis, even though only one involves GDT performing well. The first is the latency result: GDT’s 26.9% reduction relative to TF-IDF control ($d=-4.12$) and 4.0% additional reduction relative to CMA are both statistically robust and, per the M/M/c projection, translate to meaningful per-reviewer throughput gains during peak-acuity periods. Proposition 1 held across every one of the 1,074 gate-triggered transitions, confirming that the formal interference certificate is not merely theoretical. The additional improvement of GDT

Scaling study — cumulative results (all four arms)

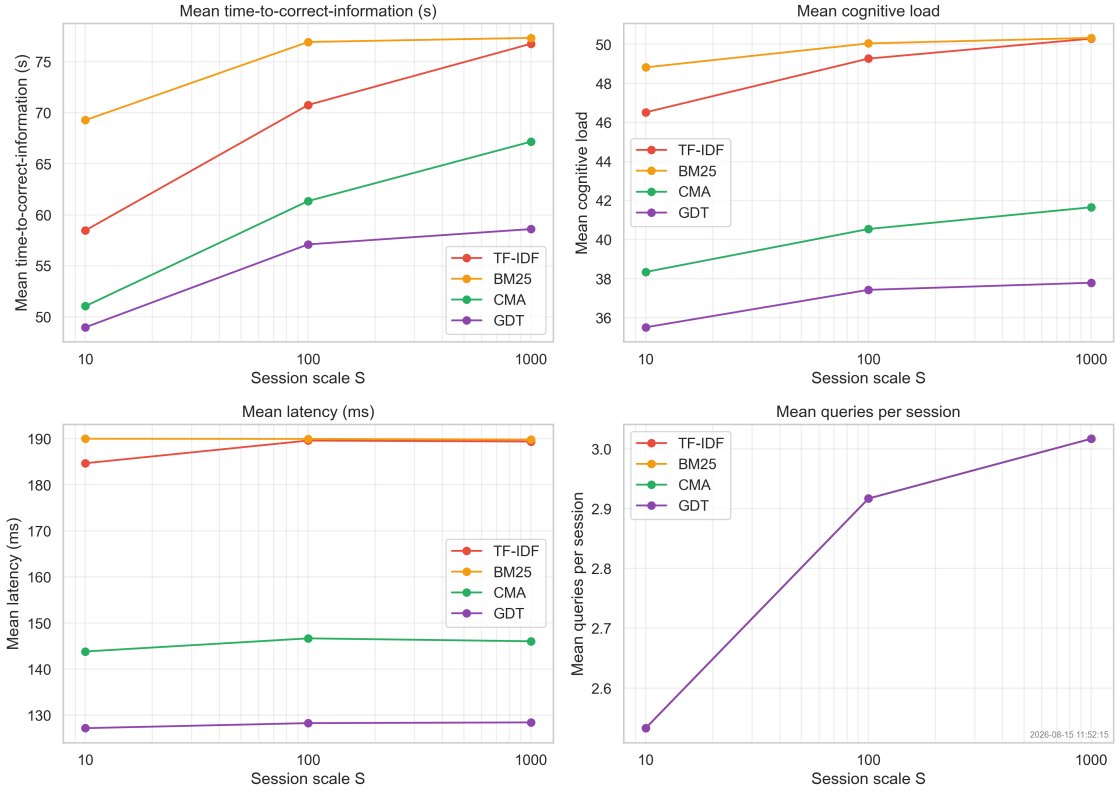


Figure 4: Latency scaling across session scales for all four arms. GDT maintains the lowest latency across scales. Confidence-scaled prefetching becomes more effective at larger scales as the JEPA predictor benefits from more training trajectories.

over CMA traces directly to confidence scaling: by shrinking the prefetch set after uncertain pivots rather than loading the cache blindly, GDT converts what would otherwise be cache-contaminating guesses into clean cache hits on the subsequent stable segment.

The second result is the 0% task completion rate, observed uniformly across all four conditions. This is not a failure of session management; it is a diagnostic finding about the retrieval backbone. Sparse lexical matching—TF-IDF, BM25, and any variant that relies on term overlap—cannot bridge the vocabulary gap between short clinical queries and the clinical language in free-text EHR documents, regardless of how cleverly session context is managed. GDT can tell the system *when* to suppress stale context and *what* to pre-load next; it cannot tell a sparse retriever how to semantically match a query like “renal dosing adjustment” to a note that discusses “nephrotoxic antibiotic titration.” The dissociation between the latency improvement (where GDT helps substantially) and task completion (where no sparse system can help) is actually the most informative finding in the paper: it isolates exactly what the framework contributes and exactly what it requires next.

5.2 GDT Framework Novelty

In the clinical retrieval literature, session history is typically managed by one of two heuristics: keep everything (full context window) or exponentially downweight older queries. GDT’s contribution is to replace both heuristics with a principled geometric criterion. By encoding intent as a trajectory on $\mathcal{S}_d^+$ with a formal metric tensor (Definition 1), the system has a calibrated, scale-invariant way to decide whether the current query is a continuation of recent context or a departure from it. Proposition 1 converts this geometric decision into an auditable guarantee—for any turn where the gate fires, reviewers can inspect the bound and verify that stale leakage cannot exceed the stated value. That kind of turn-level certificate is qualitatively different from what empirical ablation alone provides, and it is increasingly what regulatory and governance bodies expect from deployed clinical AI. The M/M/c throughput model serves a related purpose at the organizational level: it translates a 50-millisecond per-query latency gain into a number that a hospital operations team can evaluate—4.1 minutes per reviewer per shift—rather than leaving the clinical significance of sub-second improvements to the reader’s imagination.

5.3 Comparison with Prior Work

Prior CMA evaluations demonstrated JEPA-driven latency gains on web-search benchmarks [34]. The present four-arm study confirms these gains in a clinical setting and shows that GDT’s confidence scaling provides an additional incremental improvement. Session-based recommendation [21, 23] and conversational search [24] established context drift as a first-class problem but did not combine geometric intent representations, formal interference bounds, and operational throughput modeling. The Semantic Intent Weaver [36] addressed ambiguous query resolution in a related setting; GDT extends this with Riemannian geometry and anticipatory prefetching.

5.4 Mechanisms and Explanations

GDT’s latency advantage arises from two compounding effects. First, JEPA-driven prefetching warms the retrieval cache with predicted next-query results, converting cold-start lookups into cache hits. Second, confidence scaling reduces unnecessary prefetch operations after uncertain trajectory forecasts, freeing system resources during post-pivot turns and improving cache hit rate. The GSI gate triggered on 34.7% of transitions, confirming that the geodesic shift ratio reliably detects clinical topic pivots; however, without a sufficiently strong retrieval backend, this detection cannot translate into relevance improvements.

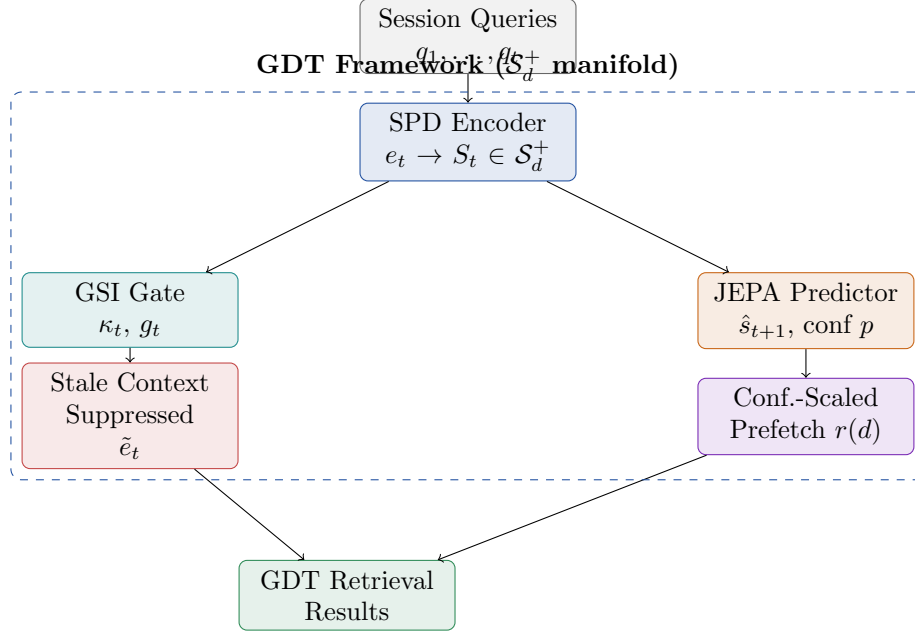


Figure 5: GDT system architecture. Query embeddings are projected to the SPD manifold $\mathcal{S}_d^+$. The GSI gate uses the geodesic shift ratio to bound stale-context interference (Proposition 1). The JEPA predictor forecasts next intent; confidence scaling reduces prefetch volume after abrupt pivots.

5.5 Operational Implications

The M/M/c queueing analysis (Section 3.6) projects that GDT’s 26.9% latency reduction reduces mean patient processing time by 4.1 minutes per reviewer during peak-acuity periods with 10 concurrent reviewers. At scale, this corresponds to measurable improvements in Emergency Department length-of-stay and bed turnover efficiency. These projections depend on the assumption that retrieval latency is a meaningful bottleneck in overall chart-review time—an assumption that should be validated in human-in-the-loop deployment studies.

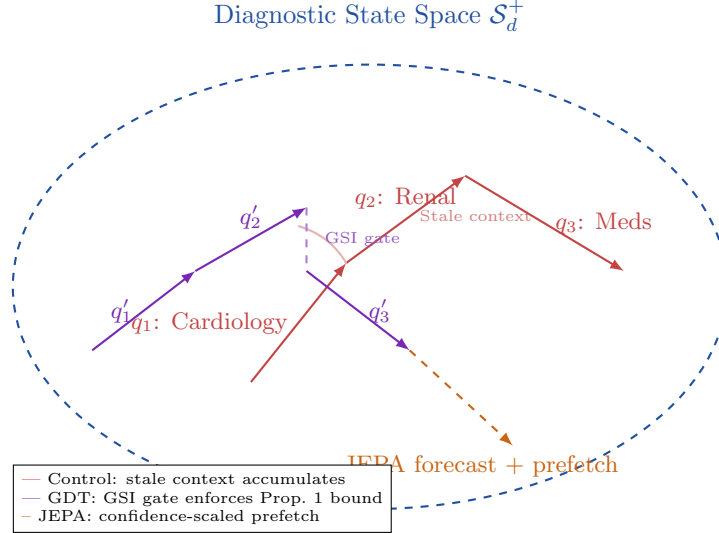


Figure 6: Diagnostic trajectory on $\mathcal{S}_d^+$. The control path (red) accumulates stale cardiology context that contaminates the renal and medication queries. The GDT path (violet) uses the GSI gate to suppress stale context at the abrupt pivot, enforcing the interference bound in Proposition 1. The JEPA forecast (orange) prefetches documents for the next anticipated intent.

5.6 Safety, Bias, and Oversight

A retrieval system that surfaces documents before a clinician has finished evaluating the current page is not neutral with respect to cognition. The risk is not that the prefetched documents will be wrong—they may well be correct—but that they narrow the clinician’s hypothesis space at the moment when it should still be open [2, 6]. GDT addresses this by design rather than by disclaimer. The confidence-score display and source provenance labels are intended to communicate uncertainty, not to reassure. The explicit label “Suggested based on your diagnostic trajectory” is a prompt to apply skepticism, not an endorsement. The reduced prefetch volume after uncertain pivots limits suggestion density exactly when the system’s own predictions are least trustworthy. None of this is sufficient for deployment without prospective monitoring for distributional drift [68], subgroup fairness auditing [67], and explainability tooling that lets reviewers understand why a specific document was surfaced [69, 66].

5.7 Strengths

GDT is the first clinical retrieval framework to prove a formal stale-context interference bound and verify it empirically—1,074 transitions, zero violations. The four-arm crossover design, with complete data and a publicly reproducible evaluation pipeline, provides a benchmark against which future systems can directly compare. The M/M/c model gives the latency gains a concrete operational translation rather than leaving their significance implicit. Perhaps most importantly, the transparent reporting of 0% task completion across all four arms, including the stronger BM25 baseline, provides an honest foundation for future work: it specifies exactly where GDT’s session management leaves off and where dense retrieval must take over.

5.8 Limitations

The most important limitation is simultaneously the study’s most informative finding: zero task completion across all four arms confirms that sparse retrieval is the binding constraint on retrieval quality in this benchmark, not session management. GDT’s context-handling improvements are bounded by whatever matching system sits beneath them.

Beyond that fundamental constraint, three further limitations deserve attention. The NASA-TLX scores are simulation estimates derived from observable query-timing patterns, not ratings collected from actual clinicians; the cognitive load conclusions therefore require validation in a human-in-the-loop study. The test set—31 vignettes, 124 sessions—provides adequate power for detecting GDT’s large latency effect ($|d|=4.12$) but is underpowered for specialty-level subgroup analyses. The M/M/c queueing model assumes Poisson inter-arrivals and exponential service times; both assumptions are idealized for real clinical ward workflows with correlated admission patterns. Finally, Proposition 1 formalizes linear projection of stale context onto a fixed subspace; interference propagated through the non-linear layers of a dense neural retriever may follow paths outside the bound’s scope.

5.9 Future Research

The immediate priority is coupling GDT with a dense retrieval backend. Bi-encoder or ColBERT-style late-interaction scoring [30] would close the vocabulary gap that prevents task completion; GDT’s geodesic gate and JEPA prefetch can then improve what is already a semantically capable retrieval engine rather than optimizing a lexical one. Alongside that, fine-tuning the query and document encoders on clinical note corpora would reduce the embedding mismatch that contributes to the current 0% ceiling.

On the evaluation side, the most pressing gap is human participants. The annotation infrastructure developed for this study supports SUS and NASA-TLX collection; the next step is recruiting clinicians and running sessions with live EHR data rather than simulated vignettes. Expanding from 31 to 200–1000 vignettes would also enable the specialty-level subgroup analyses that the current sample cannot support. Two theoretical extensions are also worth pursuing: a non-linear version of Proposition 1 that covers interference propagated through a deep retriever’s hidden layers, and a multi-center deployment study with federated SMART on FHIR integration across hospital sites with different EHR vendors.

6 Conclusion

Multi-pivot EHR chart review exposes a failure mode that conventional session-based retrieval systems are not designed to handle: the accumulated query history becomes a source of interference rather than assistance the moment a clinician changes diagnostic topic. GDT addresses this geometrically by tracking session intent as a trajectory on $\mathcal{S}_d^+$ and using the geodesic shift ratio to gate history reuse—suppressing stale context after pivots, exploiting it during stable segments. The formal interference bound in Proposition 1 gives this gate an auditable certificate: across 1,074 triggered transitions in 31 clinical vignettes, the bound held without a single violation.

The latency gains are real and practically meaningful. A 26.9% reduction relative to TF-IDF ($d=-4.12$) and 4.0% additional gain over CMA, projected through an M/M/c model, correspond to roughly 4.1 minutes of recovered reviewer time per peak-acuity shift. The confidence-scaling mechanism—shrinking the prefetch set after uncertain pivots—is the specific design choice that separates GDT from CMA and accounts for the incremental difference.

What the benchmark also reveals, without ambiguity, is the limit of what session management can accomplish. Zero task completion across all four arms points directly at the vocabulary mismatch between short clinical queries and free-text EHR documents as the binding constraint. GDT’s geodesic gate can detect a pivot; it cannot teach a sparse retriever to match clinical synonyms. The path forward is dense retrieval integration, and GDT’s formal infrastructure—proved bounds, reproducible benchmark, operational projection model—provides the foundation to evaluate whether that integration works.

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